# 人体内源代谢物在肿瘤发展和免疫调节中的研究进展

综 述

张昊仟[1,2],潘 欣[2],乔 玥[1],宣成睿[1]

[摘要] 肿瘤微环境中的代谢重塑与肿瘤的发生发展密切相关。内源代谢物不仅能够调节肿瘤形成和转移,还可通过影响免疫细胞继而削弱其抗肿瘤免疫效能。然而,当前研究多局限于单一代谢物对肿瘤的影响,并未充分考虑代谢网络的系统性、代谢产物的多样性及其对肿瘤免疫微环境的动态调控作用。深入解析内源代谢物相关代谢通路中的关键节点及其在肿瘤进展中的操纵机制,在基于干预代谢的抗肿瘤治疗策略中具有重要意义。该文系统综述了氨基酸、脂质和核苷酸等内源代谢物在肿瘤进展过程中的分子机制及其对免疫微环境的调控作用,旨在为理解代谢-免疫轴在肿瘤发生发展中的复杂网络提供新的理论基础,以期为开发靶向代谢的抗肿瘤免疫治疗策略提供参考。

[关键词] 代谢物;肿瘤微环境;氨基酸;脂质;代谢;免疫调节
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# Research progress in roles of endogenous metabolites in tumor progression and immune regulation

*ZHANG Haoqian*[1,2], *PAN Xin*[2], *QIAO Yue*[1], *XUAN Chengrui*[1*]
(1. College of Basic Medical Sciences, Inner Mongolia Medical University, Hohhot 010110, China; 2. Academy of Military Medical Sciences, Academy of Military Sciences, Beijing 100850, China)
*Corresponding author, E-mail: xuanchengruihl@163.com

[Abstract] Metabolic reprogramming in the tumor microenvironment is closely associated with tumor initiation and progression. Endogenous metabolites can not only regulate tumor formation and metastasis, but also impair antitumor immune efficacy by influencing immune cells. However, current research focuses on the effects of single metabolites on tumors rather than take into account the systemic nature of metabolic networks, the diversity of metabolic products, and their dynamic regulatory roles in the tumor immune microenvironment. An in-depth analysis of the key nodes in metabolic pathways related to endogenous metabolites and their regulatory mechanisms in tumor progression is crucial to the development of antitumor therapies based on metabolic intervention. This paper provides an overview of the molecular mechanisms by which endogenous metabolites, such as amino acids, lipids and nucleotides, contribute to tumor progression and regulate the immune microenvironment. By offering new theoretical insights into the complex network of the metabolism-immune axis in tumor development, this review aims to provide data for the development of metabolism-targeted antitumor immunotherapy strategies.

[Key words] metabolites; tumor microenvironment; amino acid; lipid; metabolism; immunomodulation

代谢重编程作为肿瘤的重要特征之一,已成为近年来的研究热点。随着高通量代谢组学和系统生物学的不断发展,研究人员对内源代谢物在肿瘤发生、发展及免疫逃逸中的复杂作用有了更深入的理解。内源代谢物不仅是代谢过程的副产物,还作为关键信号分子参与肿瘤微环境(tumor microenvironment,TME)的调控。

本文对氨基酸、脂质和核苷酸等3类内源代谢物在肿瘤发生、发展及免疫调控中的研究进展进行综述,并深入探讨其在肿瘤生物学中的相互作用网络,旨在为新型肿瘤诊疗策略的开发提供理论依据,同时为应对肿瘤的异质性和治疗耐药性等问题提出新的思路。

[作者简介] 张昊仟,男,硕士研究生,研究方向:基础病理学研究,Tel:13621380213,E-mail:2022110033@stu.immu.edu.cn
[作者单位] 1.内蒙古医科大学基础医学院,呼和浩特 010110;2.军事科学院军事医学研究院,北京 100850
[通信作者] 宣成睿,Tel:18647143915,E-mail:xuanchengruihl@163.com

## 1 内源代谢物与肿瘤微环境

TME具有高度复杂性和显著异质性，由肿瘤细胞、基质细胞、免疫细胞、血管和细胞外基质等多种成分组成，在肿瘤发生、发展和转移中发挥关键作用[1]。近年来，内源代谢物在TME中的调节作用备受关注。研究表明，随着代谢组学技术的进步，内源代谢物不仅在细胞内发挥代谢功能，还通过调节TME中各种细胞的活动，间接影响肿瘤的生长和转移[2]。

肿瘤代谢重编程是肿瘤的重要特征，肿瘤细胞通过调控代谢途径以满足其快速增殖的需求[3,4]。经典的Warburg效应表明，即便在有氧条件下，肿瘤细胞依然依赖糖酵解获取能量。此外，肿瘤细胞通过谷氨酰胺代谢和脂质代谢等途径，增强其代谢灵活性和生存能力[5]。进一步研究表明，这些代谢改变通过复杂的代谢网络和表观遗传调控重塑TME中各类组分的功能状态。在基质细胞中，活化的肿瘤相关成纤维细胞(cancer-associated fibroblasts，CAF)经历显著的代谢重塑，这一过程主要受低氧诱导因子-1α/丙酮酸脱氢酶激酶1(hypoxia induced factor-1α/pyruvate dehydrogenase kinase 1，HIF-1α/PDK1)介导的糖酵解增强和c-Myc调控的谷氨酰胺代谢改变驱动[6]。活化的CAF分泌乳酸、谷氨酰胺、酮体等多种代谢产物，这些代谢物通过代谢-表观遗传轴促进肿瘤进展[3]。在血管重构中，TME积聚的代谢产物通过多层次机制调控血管生成。例如，乳酸通过单羧酸转运蛋白1/4(monocarboxylate transporter 1/4，MCT1/4)介导的pH值改变激活血管内皮生长因子(vascular endothelial growth factor，VEGF)信号通路，而前列腺素E2(prostaglandin E2，PGE2)则通过EP3/EP4受体激活磷脂酰肌醇3激酶/蛋白激酶B(phosphatidylinositol 3-kinase/protein kinase B，PI3K/AKT)和细胞外调节蛋白激酶(extracellular regulated protein kinases，ERK)级联信号，上调血管生成因子的表达[7,8]。在细胞外基质重塑过程中，内源代谢物主要通过3个方面发挥作用：(1)乳酸诱导的酸性微环境激活基质金属蛋白酶(matrix metalloproteinase，MMP)，特别是MMP2/9；(2)谷氨酰胺代谢产物α-酮戊二酸参与胶原交联修饰；(3)脂质过氧化产物通过花生四烯酸5-脂氧合酶(arachidonate 5-lipoxygenase，ALOX5)通路调控基质硬度[9,10]。这些改变最终导致细胞外基质的生物力学特性发生显著变化。在免疫调节网络中，内源代谢物通过吲哚胺2，3-双加氧酶1/色氨酸2，3-双加氧酶(indoleamine 2，3-dioxygenase 1/tryptophan-2，3-dioxygenase，IDO1/TDO)-犬尿氨酸-一般性调控阻遏蛋白激酶2(general control nonderepressible 2，GCN2)/哺乳动物雷帕霉素靶蛋白(mammalian target of rapamycin，mTOR)轴抑制T细胞功能，并通过PGE2-EP2/4-腺苷-3′，5′-环化-磷酸(cyclic adenosine monophosphate，cAMP)信号通路抑制自然杀伤细胞(natural killer cells，NK)和$CD8^+$ T细胞活性。同时，内源代谢物还通过CD39/CD73-腺苷-A2AR/A2BR信号通路构建免疫抑制网络[11,12]。此外，内源代谢物可通过诱导组蛋白乳酰化、β-羟基丁酰化等表观遗传修饰，调控免疫细胞的基因表达谱[4]。

值得注意的是，这些代谢网络并非静态存在，而是形成复杂的反馈调节环路。例如，CAF分泌的代谢产物可诱导T细胞发生表观遗传修饰，而活化的免疫细胞又可通过分泌细胞因子反向调控基质细胞代谢[5,13]。这种动态的代谢-免疫-基质相互作用网络，成为推动肿瘤免疫逃逸和进展的关键机制。

## 2 关键内源代谢物与肿瘤的相互作用

在肿瘤生物学中，不同类型的内源代谢物在肿瘤进展、免疫调节和治疗反应中起着关键作用。氨基酸、脂质和核苷酸代谢产物作为当前研究的重点，通过复杂的分子机制调控肿瘤细胞及其微环境中的多种功能。此外，这些代谢物为肿瘤细胞的快速增殖提供能量和构建材料，并通过调控微环境中免疫细胞和基质细胞的功能，促进肿瘤的侵袭和扩散。

### 2.1 氨基酸代谢物

氨基酸在肿瘤生物学事件中具有关键作用，除为肿瘤细胞的生长提供必要物质外，还可通过其代谢产物进一步影响TME和免疫反应[14]。其中，色氨酸、精氨酸和支链氨基酸(branched-chain amino acids，BCAA)的代谢尤为重要，它们通过复杂的分子机制调控肿瘤进展和免疫逃逸。

**2.1.1** *色氨酸代谢通路及其在肿瘤免疫中的作用* 色氨酸代谢通路，尤其是由IDO1和TDO介导的犬尿氨酸途径，在肿瘤免疫逃逸中发挥关键作用[15]。该通路的激活导致TME中色氨酸的耗竭及犬尿氨酸及其代谢产物的积累，形成免疫抑制性环境[16]。IDO1的表达受多种炎症因子调控，包括干扰素γ、肿瘤坏死因子α和白细胞介素6，并通过信号传导和转录激活因子3(signal transducer and activator of transcription 3，STAT3)和核因子κB(nuclear factor kappa-B，NF-κB)形成正反馈环路，导致了某些肿瘤中IDO1表达的持续升高[17]。

IDO1/TDO催化色氨酸转化为N-甲酰基犬尿氨酸，随后生成犬尿氨酸、喹啉酸和吡啶甲酸等代谢物[18]，这些代谢物通过多种机制抑制T细胞功能：色氨酸耗竭激活GCN2通路，引发未折叠蛋白反应，抑制T细胞增殖并诱导其凋亡[11]；犬尿氨酸及其代谢物作为芳香烃受体（aryl hydrocarbon receptor，AhR）的配体，增强调节性T细胞（regulatory T cells，Treg）功能[19]。AhR还与V-Rel网状内皮增生病毒癌基因同源物B（v-rel reticuloendotheliosis viral oncogene homolog B，RelB）协同调控细胞程序性死亡-配体1（programmed cell death ligand 1，PD-L1）的表达，揭示了代谢与免疫检查点的关联[20]。此外，这些代谢物通过抑制mTOR复合物1（mTOR complex 1，mTORC1）活性，进一步影响T细胞的代谢与功能[12]。除酶活性外，IDO1还具有非经典的信号转导功能，即通过招募蛋白酪氨酸磷酸酶1（SH2-containing protein tyrosine phosphatase 1，SHP1）和SHP2参与转化生长因子β（transforming growth factor-β，TGF-β）的长期诱导，从而维持免疫耐受状态[21]。

**2.1.2** *精氨酸代谢与肿瘤细胞的生存调节* 精氨酸代谢在肿瘤生物学中具有双重作用，主要通过一氧化氮合酶（nitricoxide synthase，NOS）和精氨酸酶（arginase，ARG）2条途径发挥作用[22]。NOS介导的低浓度一氧化氮通过HIF-1α和VEGF等S-亚硝基化修饰关键蛋白促进血管生成和细胞存活[7]；而高浓度一氧化氮则通过诱导DNA损伤并激活p53信号通路，导致细胞凋亡[23]。一氧化氮还通过PDK等硝基化修饰代谢酶进而影响肿瘤能量代谢，并通过抑制T细胞受体信号传导和上调PD-L1表达促进免疫抑制微环境的形成[6]。

ARG代谢生成的多胺（如精胺和亚精胺）在肿瘤细胞增殖和存活中起重要作用。多胺不仅能影响蛋白质的翻译后修饰[如超精胺化对真核生物翻译起始因子5A（eukaryotic translation initiation factor 5A，eIF5A）活化至关重要]，还可通过调节染色质结构和基因表达影响代谢酶，如M2-型丙酮酸激酶（pyruvate kinase isozyme typeM2，PKM2），继而参与Warburg效应[24]。

**2.1.3** *支链氨基酸代谢对肿瘤生长与免疫微环境的影响* BCAA在肿瘤代谢中具有独特作用，既可作为蛋白质合成的原料，也可与Sestrin1/2蛋白（SESN1/2）结合，激活mTORC1信号通路[25]。这一过程涉及复杂的蛋白网络调控，如SAMTOR作为甲硫氨酸感受器的作用，以及KICKSTOR调节因子在整合氨基酸信号中的角色。胰腺导管腺癌中，胰腺癌细胞的BCAA代谢增强，不仅为其提供了能量和合成前体，还提高了其对氧化应激的耐受性[26]。

BCAA的代谢能够影响肿瘤免疫微环境。肿瘤细胞高表达支链氨基酸转氨酶1（branched chain aminotransferase 1，BCAT1），导致免疫微环境中BCAA耗竭[27]，抑制免疫微环境中的T细胞功能和增殖活性，以维持肿瘤细胞内能量代谢平衡、促进肿瘤细胞在恶劣环境中的生长和侵袭[28]。转移性乳腺癌细胞通过增强BCAA的摄取和氧化，支持其在转移部位的存活和生长[29]。

### 2.2 脂质代谢物

脂质代谢重编程是肿瘤细胞适应微环境应激并满足快速增殖需求的关键策略之一[9]。在众多脂质代谢物中，PGE2、鞘脂类代谢物和胆固醇代谢产物在肿瘤进展和免疫逃逸中发挥着重要作用。这些代谢物通过复杂的信号网络调控肿瘤细胞的生存、增殖、侵袭和转移能力，并显著影响TME中免疫细胞的功能[10]。

**2.2.1** *PGE2与肿瘤发生及免疫调节* PGE2是花生四烯酸代谢的主要产物之一，通过多种机制参与肿瘤的发生、发展和免疫逃逸[30]。其合成由环氧化酶（cyclooxygenase，COX）和前列腺素E2合酶（prostaglandin E2 synthase，PGES）共同催化，COX-2在多种肿瘤中高表达[31]。PGE2通过与4种G蛋白偶联受体（EP1-4）结合，激活多条下游信号通路，如cAMP/蛋白激酶A（protein kinase A，PKA）通路、PI3K/AKT通路和丝裂原活化蛋白激酶（mitogen-activated protein kinase，MAPK）通路，以促进肿瘤细胞增殖、抑制凋亡，并增强侵袭和转移能力[8]。此外，PGE2还能通过激活β连环蛋白（β-catenin）信号通路促进肿瘤干细胞的自我更新[32]。

在免疫调节方面，PGE2通过cAMP-PKA信号通路抑制NK细胞和CD8$^+$T细胞的细胞毒性，同时促进髓系来源抑制细胞（myeloid-derived suppressor cells，MDSC）的募集和活化，构建免疫抑制网络[33]。PGE2还通过上调PD-L1的表达，进一步增强免疫抑制作用[34]。这一机制为联合靶向COX-2/PGE2轴与PD-1/PD-L1通路的治疗提供了理论依据[35]。

**2.2.2** *鞘脂类代谢在肿瘤细胞存活与免疫中的作用* 鞘脂类代谢物，如神经酰胺和鞘氨醇-1-磷酸（sphingosine-1-phosphate，S1P），在细胞的存活、增殖和迁移中发挥关键作用[36]。神经酰胺促进细胞凋亡，S1P则促进细胞生存与迁移[37]。肿瘤细胞通过上调鞘氨醇激酶1（sphingosine kinase 1，SPHK1）增加S1P的生成，继而激活PI3K/AKT和ERK等信号通路，促进肿瘤细胞的增殖与迁移[38]。此外，鞘氨醇-1-磷酸/1-磷酸鞘氨醇受体1（sphingosine-1-phosphate/

sphingosine-1-phosphate receptor 1,S1P-S1PR1)轴在调节肿瘤相关炎症和免疫抑制中也发挥重要作用,维持Treg细胞的稳定性并抑制效应T细胞的活化[39]。SPHK1的高表达与多种化疗药物的耐药性相关,靶向S1P受体及鞘氨醇激酶的策略展现出抗肿瘤潜力[40]。

**2.2.3** *胆固醇代谢异常与肿瘤进展* 胆固醇代谢异常与癌症的发生和进展密切相关[41]。肿瘤细胞通常表现出胆固醇合成和摄取增加,不仅支持细胞膜的合成,还为信号转导提供调节分子[42]。胆固醇调节元件结合蛋白(sterol-regulatory element binding proteins,SREBP)转录子的异常激活促进胆固醇合成,支持肿瘤的快速增殖[43]。氧化胆固醇,如27-羟基胆固醇,通过激活肝X受体(liver X receptor,LXR)信号通路促进肿瘤转移[44]。肿瘤细胞还可上调胆固醇转运体三磷酸腺苷结合盒转运体A1(ATP binding cassette transporter Al,ABCA1)的表达,降低胞膜胆固醇含量,从而抑制$CD8^+$T细胞的杀伤功能[45]。

基于脂质代谢在肿瘤生物学中的作用,研究者开发了多种靶向治疗策略。除传统他汀类药物外,靶向SREBP通路的小分子抑制剂也展现出抗肿瘤潜力[46]。如何在发挥抗肿瘤作用的同时最小化对正常组织的影响,仍是当前面临的挑战[47]。

### 2.3 核苷酸代谢物

核苷酸代谢不仅为肿瘤细胞快速增殖提供必要的"原料",其代谢产物还通过多种信号通路调节肿瘤微环境的组成和免疫系统的反应,影响肿瘤生长和免疫逃逸[48]。在众多核苷酸代谢物中,腺苷和cAMP在肿瘤代谢调控中发挥着尤为重要的作用。

**2.3.1** *腺苷在TME中的免疫逃逸机制* 腺苷作为强效的免疫抑制因子,在肿瘤免疫逃逸中起到核心作用[49]。在低氧和营养匮乏的TME中,胞外核苷酸水解酶CD39和CD73的表达显著上调,导致细胞外ATP被快速水解为腺苷[50]。腺苷通过与G蛋白偶联受体A1R、A2AR、A2BR和A3R结合,尤其是A2AR和A2BR,在免疫调节中发挥重要作用。腺苷通过激活cAMP-PKA信号通路,抑制效应T细胞和NK细胞的活化及功能,减少细胞因子的生成与细胞毒性分子的释放[51]。此外,腺苷还能增强Treg的分化和功能进而加剧免疫抑制[52],以及通过A2BR信号通路促进MDSC的扩增和活化[53]。

研究发现,某些肿瘤细胞在葡萄糖缺乏时能够利用腺苷作为能量底物和生物合成前体[54]。通过A3R信号通路,腺苷能够促进某些类型肿瘤细胞的上皮-间质转化和侵袭能力[55]。

**2.3.2** *cAMP在肿瘤细胞代谢和免疫调节中的作用* cAMP作为关键的第二信使[56],其水平由腺苷酸环化酶和磷酸二酯酶精确调控。cAMP信号失调与肿瘤细胞的恶性特征密切相关,包括无限增殖、抗凋亡及增强的转移能力[57]。cAMP-PKA信号轴通过激活cAMP效应元件结合蛋白转录因子,上调代谢相关基因如己糖激酶2(hexokinase 2,HK2)和丙酮酸激酶M2(pyruvate kinase M2,PKM2)从而促进Warburg效应[58]。同时,通过磷酸化Yes相关蛋白(Yes-associated protein,YAP)调控Hippo信号通路,cAMP-PKA信号轴也可直接影响肿瘤细胞生长和组织及器官的整体大小[59]。

在免疫调节中,cAMP通过部分减少白细胞介素2(interleukin-2,IL-2)生成和CD25表达抑制T细胞活化和增殖[60],并通过抑制NF-κB和活化T细胞核因子(nuclear factors of activated T cells,NFAT)信号通路进而削弱T细胞效应功能[61]。免疫检查点如细胞毒T淋巴细胞相关抗原4(cytotoxic t lymphocyte-associated antigen-4,CTLA-4)可提升T细胞内cAMP水平以发挥免疫抑制作用[62]。

基于这些发现,CD39和CD73抑制剂、腺苷受体拮抗剂(特别是A2AR拮抗剂)等多种靶向治疗策略正在开发并进入临床试验,联合免疫检查点抑制剂的初步结果显示出抗肿瘤潜力[63]。

## 3 总结与展望

内源代谢物在肿瘤进展和免疫调节中的作用已成为近年来癌症研究的重点。深入阐明其复杂的作用机制,为理解代谢与癌症的关系提供了重要基础。然而,目前的研究多聚焦于单一代谢物的直接影响,未充分考虑代谢网络的整体性和动态性。因此,未来的研究应整合多组学分析,全面揭示代谢物-肿瘤轴的动态调控网络。

开发针对特定代谢通路或微生物群的高特异性调节剂,以提高治疗的精准性,是当前研究的重要方向。深入研究内源代谢物在肿瘤中的作用机制,有望为肿瘤的早期诊断、精准治疗和预后评估提供新的思路,最终改善患者的治疗效果和生存质量。

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